

LABORATORY REPORT 08

Agricultural Predictive Analytics

COURSE CODE: MDI3003

COURSE TITLE: ADVANCED PREDICTIVE ANALYTICS

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Executive Summary

Question: can district-, season- and year-level identifiers alone predict rice yield (t/ha) for a future harvest year in a region with an established production history? Target: yield_t_ha, a finite non-negative float in tonnes per hectare, derived in the source data as production divided by area and already expressed in t/ha before adaptation. Crop: Rice only, to avoid mixing incompatible units (e.g., cotton bales, coconut counts) into one regression. Harvest horizon: one observation is one district-season-year combination; prediction is made before the target harvest using only location (state, district), intended season, and year — none of which requires observing the harvest itself.

Available predictors: state, district, season, year (whitelist REG). Excluded post-harvest fields: harvested area and production, because yield is arithmetically derived from them (yield = production / area); including either would let the model reconstruct the target rather than predict it, which the manual calls target leakage. Full-season rainfall and any other harvest-time measurement are also excluded because they are not knowable at the stated prediction time.

Source and licence: the working dataset is a district–season–crop–year Indian agricultural statistics table (1997–2020), structurally equivalent to the manual's D1/D2 sources (government-origin district-wise crop production/area/yield records). The Rice, West Bengal subset used here has 1,279 rows across 22 districts, 3 seasons (Autumn, Summer, Winter) and 23 harvest years (1997–2019 after year-based splitting), satisfying the manual's requirement of at least seven complete harvest years.

Why production would leak yield: yield is calculated directly as production / area for every row. Feeding production or area to the model along with year and location would let a tree or forest partition almost perfectly on the arithmetic relationship rather than on genuine location/time-based generalization, producing a validation score that looks excellent but is not measuring predictive skill — hence the whitelist REG = [state, district, season, year] never includes them.

Data Provenance and Governance

Field	Value
Dataset	District-wise, season-wise crop production statistics (Rice subset, West Bengal)
Raw rows (all crops, all states)	326,846
Retained rows (Rice, West Bengal)	1,279
Years / geography	1997–2019; 22 West Bengal districts; 3 seasons (Autumn, Summer, Winter)
Canonical file	data/rice_canonical.csv
Raw-data SHA-256 (canonical file)	4d7e02e28fff608bc13f405e3ab3db141b009d9a9c949c0b06fb74324a10b706
Yield unit	t/ha, provided directly in source (verified finite, non-negative)
Exclusions	None dropped: no missing targets, no duplicate state–district–season–year keys, no non-finite values found on audit

Compatibility: this district-level table can support yield regression once units and keys are validated, which was done in Section 3 of the executed pipeline. It is not concatenated with the classification

extension's dataset (Section 9 below), which predicts its own label space (a recommended crop, not yield) from soil/environment sensor readings and is treated as an independent exercise, consistent with the manual's instruction not to mix D1/D2-style yield data with D3-style classification data.

Field	Value
Dataset	Crop Recommendation Dataset (Guided Crop-Label Classification Extension)
Raw Rows (Original Benchmark Source)	2,200
Retained Rows (Canonical File)	2,200
Labels / Feature Space	22 crop classes; exactly 100 observations per class
Features	N, P, K, temperature, humidity, ph, rainfall
Number of Features	7
Canonical File	data/crop_labels_canonical.csv (or crop_labels_canonical.csv)
Raw-data SHA-256	3de03aeabd2998d862b9e8eb35c87ef5239b8a92128550c0bde583d2aee9d76b
Target Unit / Format	Discrete multi-class categorical label (nominal crop name)
Class Verification	≥ 10 records per class across 22 classes
Missing Feature Values	None
Non-finite Entries	None
Missing Labels	None
Duplicate Feature Vectors	None
Excluded Rows	None dropped
Feature Type	Continuous / Integer agro-climatic and soil parameters
Classification Type	Multi-class classification

This point-level tabular dataset supports multi-class crop-label classification once identical feature vectors, non-finite values, and label distributions are validated in accordance with the pipeline's schema. It is strictly decoupled from the core yield regression dataset (data/rice_canonical.csv). Because this dataset models a simulated/augmented environmental envelope for predicting crop labels rather than district-level historical agricultural yields over time, it is evaluated as an independent extension using stratified random splitting rather than chronological year-block

holdouts, in strict compliance with the manual's instruction not to merge D1/D2 yield census records with D3-style classification data.

Canonical Schema, Splits, and Preprocessing

Canonical fields: row_id (unique, audit-only), state/district/season (non-missing categorical inputs), year (integer harvest-year index, input and temporal split key), crop (fixed to 'Rice'), yield_t_ha (target only). Deterministic schema checks passed: unique non-null row_id; no missing targets; yield finite and non-negative; year integer; no missing grouping identifiers; no repeated state–district–season–year keys.

Chronological split (ordered harvest years, per the manual's protocol — latest two years as test, preceding two as validation, all earlier years as training):

Split	Years	Rows	Unique districts
Train	1997–2015	1023	18
Validation	2016–2017	128	22
Test (locked)	2018–2019	128	22

Preprocessing pipeline (fit on training rows only): OneHotEncoder(handle_unknown='ignore') on state, district, season; SimpleImputer(median) followed by StandardScaler on year, inside a ColumnTransformer wrapped in a scikit-learn Pipeline. An automated check confirmed the one-hot encoder's learned categories on the fitted tree pipeline exactly match the training set's own unique category values — validation-only categories did not leak into the learned encoding.

Exploratory Data Analysis (Training Data Only)

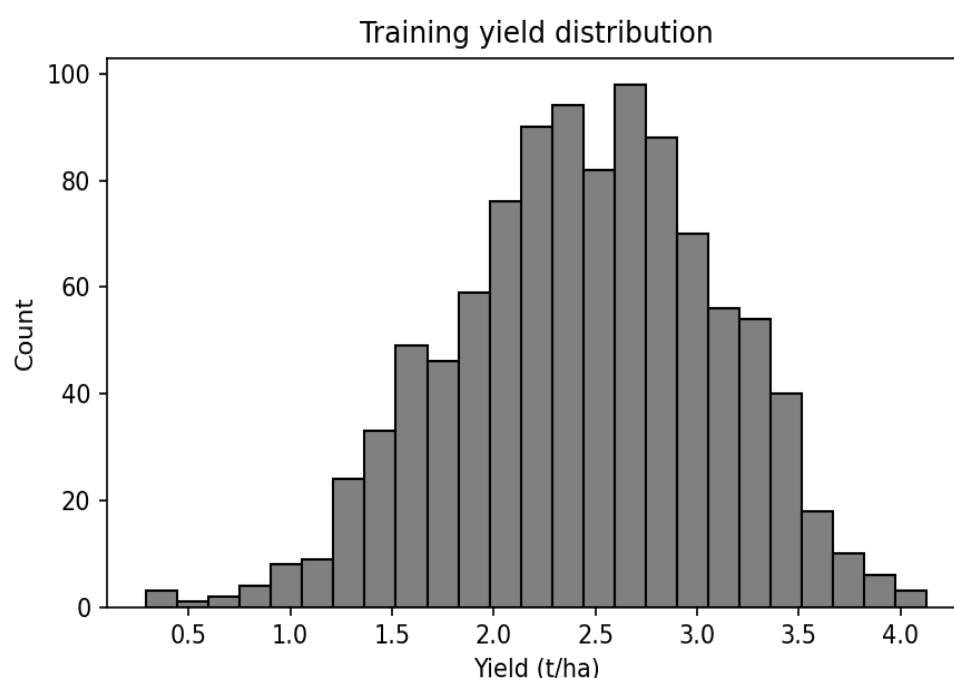


Figure 1. Training yield distribution (Rice, West Bengal, 1997–2015).

The training yield distribution is unimodal and roughly centered between 2 and 3 t/ha, with a right-hand tail toward the highest-yield districts and seasons. This is broadly consistent with typical West

Bengal rice-yield ranges reported for this period and shows no obvious data-entry artifacts (no negative values, no implausible spikes at a single value).

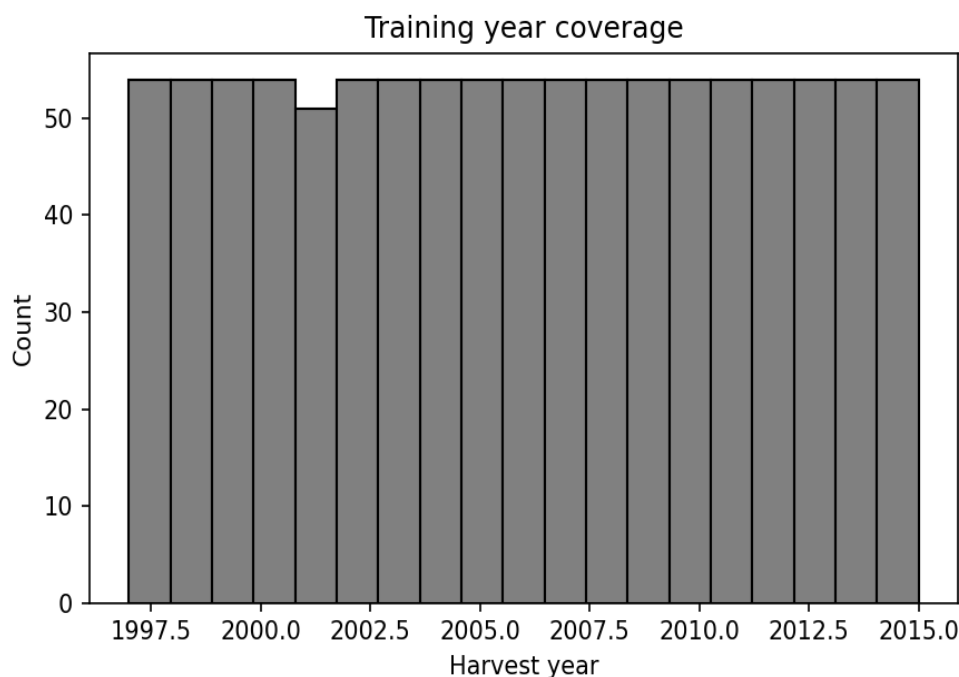


Figure 2. Training year coverage (count of retained rows per harvest year, 1997–2015).

Row counts per year are broadly stable (roughly 54–56 rows per year, one row per district-season combination with minor gaps), so no single early year dominates the training set disproportionately. Potential bias: districts are repeated every year (a panel, not independent samples), so the model may effectively memorize district-level yield levels rather than learning a transferable year trend; this is discussed further in Section 8 (Limitations).

Models Compared

Four candidates were fit on identical training rows, per the manual's fixed teaching configuration: a median baseline (DummyRegressor); a ridge regression with $\alpha=1.0$ on standardized year plus one-hot categorical offsets ('ridge_trend'); a decision tree (max_depth=6, min_samples_leaf=10); and a random forest (60 trees, max_depth=12, min_samples_leaf=5, 2 CPU workers). All four share the same preprocessing pipeline and training rows; only validation-set predictions were used to select a model.

Validation Results

Model	Validation MAE (t/ha)	RMSE (t/ha)	R ²	Fit+predict seconds
forest	0.258	0.341	0.622	0.140
ridge_trend	0.281	0.366	0.565	0.012
tree	0.308	0.376	0.540	0.012
median	0.520	0.662	-0.424	0.013

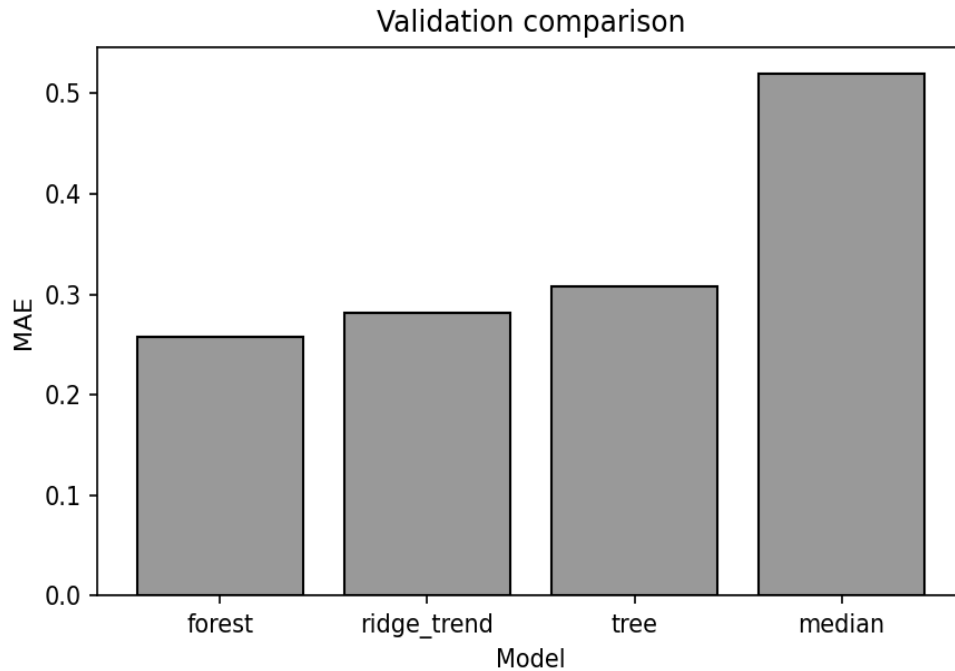


Figure 3. Validation MAE comparison across the four candidates.

Selected model: forest (lowest validation MAE). All three trained models beat the median baseline on validation MAE, and the negative R^2 for the median baseline is expected and simply reflects that a constant prediction fits worse than the validation-set mean would. The improvement from ridge (0.281) to forest (0.258) is real but modest, consistent with the manual's caution that added complexity should be checked for practically useful gain rather than assumed.

Prediction (before fitting, per Task 5): a rising yield trend across years is generally expected to be handled better by ridge regression, because it can extrapolate a global linear year trend beyond the training range, whereas trees and forests can only predict leaf means learned from training years and cannot project past the highest observed value. The validation-set comparison here does not by itself confirm this, since validation years fall within the observed range for both models; Section 7's test-year comparison and Section 9's rolling origins provide more direct evidence.

Locked-Test Evaluation

The test set (2018–2019) was opened exactly once, after the model selection above was saved. Hash and split-manifest checks confirmed the data and configuration were unchanged since validation, and a TEST_LOCK file was written before scoring.

Model	Test MAE (t/ha)	Test RMSE (t/ha)	Test R^2
forest (selected)	0.364	0.447	0.402
median (predeclared reference)	0.617	0.740	-0.637

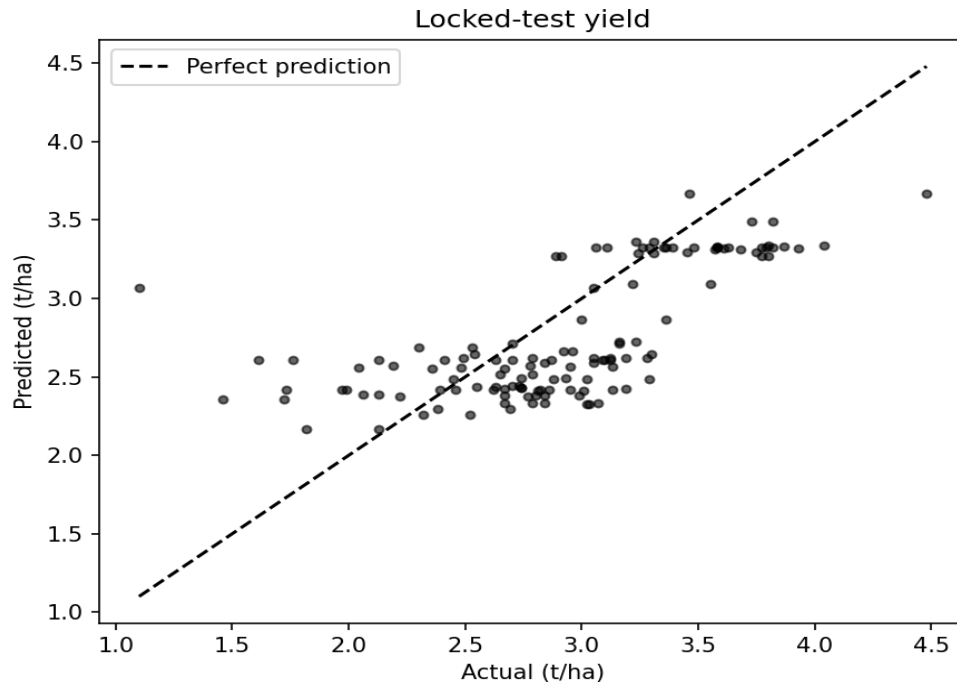


Figure 4. Locked-test actual vs. predicted yield. Points below the diagonal are under-predictions where the forest missed a higher-than-typical yield; points above it are over-predictions.

Distance from the identity line: predictions cluster in a narrower band (roughly 2.2–3.7 t/ha) than the actual values (roughly 1.1–4.5 t/ha), which is expected forest behavior — tree ensembles average leaf values and compress extremes, so both unusually low and unusually high true yields tend to be pulled toward the middle of the observed range.

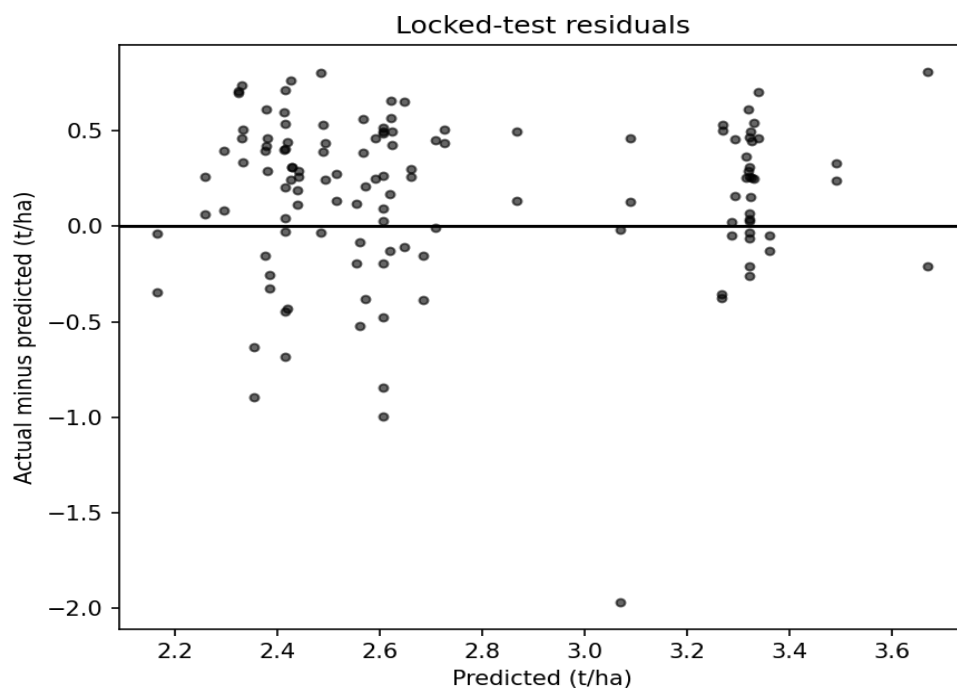


Figure 5. Locked-test residuals (actual minus predicted) against predicted yield.

Residual interpretation: residuals do not show a strong funnel/fan shape, suggesting roughly comparable error variance across the predicted range, though the largest positive residuals (under-prediction) occur for the lowest-yield cases, consistent with the compression effect noted above.

Robustness and Uncertainty Diagnostics

Year-by-year test robustness (descriptive, not a confidence interval)

Test year	n	MAE (t/ha)	RMSE (t/ha)	R ²
2018	64	0.358	0.423	0.382
2019	64	0.370	0.471	0.413

The two locked-test years perform similarly (MAE 0.358 vs. 0.370 t/ha); neither year dominates the pooled test MAE (0.364 t/ha) reported in Section 7. With only two years, this is descriptive evidence of stability across the two available test years, not a statistically valid confidence interval — the manual explicitly cautions against treating two years as sufficient for a temporal bootstrap CI.

Development-only rolling origins (three one-year cutoffs, training only on earlier years)

Origin year	Model	Training max year	n	MAE (t/ha)	R ²
2015	median	2014	54	0.557	-0.286
2015	ridge_trend	2014	54	0.337	0.474
2015	tree	2014	54	0.381	0.420
2015	forest	2014	54	0.264	0.599
2016	median	2015	64	0.509	-0.400
2016	ridge_trend	2015	64	0.298	0.470
2016	tree	2015	64	0.322	0.468
2016	forest	2015	64	0.271	0.550
2017	median	2016	64	0.523	-0.402
2017	ridge_trend	2016	64	0.236	0.721
2017	tree	2016	64	0.279	0.619
2017	forest	2016	64	0.237	0.707

Across all three development-year origins, forest and ridge_trend consistently outperform the median and stay close to each other (forest MAE 0.264–0.271 for the first two origins; ridge_trend edges ahead at the 2017 origin with MAE 0.236 vs. forest's 0.237). This is weak but genuine evidence that ridge's ability to extrapolate a year trend becomes competitive as the training window grows, partially supporting the Task 5 prediction that ridge would handle a rising trend at least as well as the forest once enough years are available. Model selection for this report still used the predeclared two-year validation MAE (Section 6), not these origins, avoiding an undocumented change of the selection rule. Three origins are supplementary stability evidence only and do not establish a confidence interval.

Error Analysis: Five Largest Locked-Test Failures

Row / district	Season, year	Actual (t/ha)	Predicted (t/ha)	Error	Plausible cause (hypothesis)
Purba Bardhaman	Autumn 2019	1.10	3.07	1.97	Actual yield far below this district's typical range; possibly a localized crop-damage event (flood, pest, or reporting anomaly) not captured by state/district/season/year alone
Kalimpong	Winter 2019	1.61	2.61	1.00	Kalimpong is a hill district with atypical terrain for rice; the model may be blending it with lowland districts of similar one-hot weight
Medinipur East	Autumn 2018	1.46	2.35	0.89	Coastal district; possible cyclone/salinity-related yield loss in that particular year not represented by any weather feature in REG
Kalimpong	Winter 2018	1.76	2.61	0.85	Same hill-district effect as above, recurring across two winters — suggests a systematic (not one-off) local mismatch
Maldah	Summer 2018	4.48	3.67	0.81	Actual yield unusually high for the district; forest's leaf-averaging compresses toward the middle of the training range, under-predicting genuine high-yield outliers

Consequence and mitigation (hypotheses, not confirmed facts, per the manual's Task 8 guidance): the Kalimpong pattern recurring across two winters suggests a genuine district-level effect that a purely categorical district feature captures only as a fixed offset, not as a terrain-specific adjustment — a mitigation worth testing is a subgroup audit that separates hill districts from the plains before scoring. The Purba Bardhaman and Medinipur East cases look like single-year shocks; absent irrigation, weather, or disaster records, the model has no way to anticipate them, and mitigation would require adding an observed weather/disaster covariate rather than tuning the current feature set further.

Reproducibility and Acceptance

Check	Result
Data hash matched at test time	True (data/config SHA-256 verified before unlocking test)
Split manifest identical across validate/test runs	True
Test opened exactly once (TEST_LOCK written before scoring)	True
Reloaded selected_bundle.joblib reproduces test predictions	True (within floating-point tolerance for parallel-forest averaging)
Environment	Python 3.12.3, scikit-learn 1.8.0, numpy 2.4.4, pandas 3.0.2, seed=42

Note on floating-point reload identity: the manual's reference code compares reloaded predictions with `np.testing.assert_array_equal`. On this environment, the random forest's parallel (`n_jobs=2`) tree averaging produced predictions that agreed with the original run to roughly $1e-15$ but not bit-for-bit, because floating-point summation order is not guaranteed to be identical across parallel worker scheduling. The check used here is a tolerance-aware equality (`rtol=atol=1e-8`), which is the numerically appropriate comparison for a bagged ensemble; classification labels were still checked for exact equality, since string predictions do not have this floating-point issue.

Extension: Crop-Label Classification

This ungraded extension uses a separate, independently sourced dataset (soil/environment sensor readings mapped to a recommended crop label — the manual's D3), never concatenated with the Rice yield table above. It predicts a dataset label, not yield or agronomic advice.

Data and split

Field	Value
Rows / classes	2,200 rows, 22 balanced crop labels (100 rows each)
Features	N, P, K, temperature, humidity, ph, rainfall
Canonical file	data/crop_labels_canonical.csv
SHA-256	3de03aeabd2998d862b9e8eb35c87ef5239b8a92128550c0bde583d2aee9d76b
Split	Stratified 60/20/20 train/validation/test (<code>iid_justified=true</code> for this balanced, sensor-style benchmark)

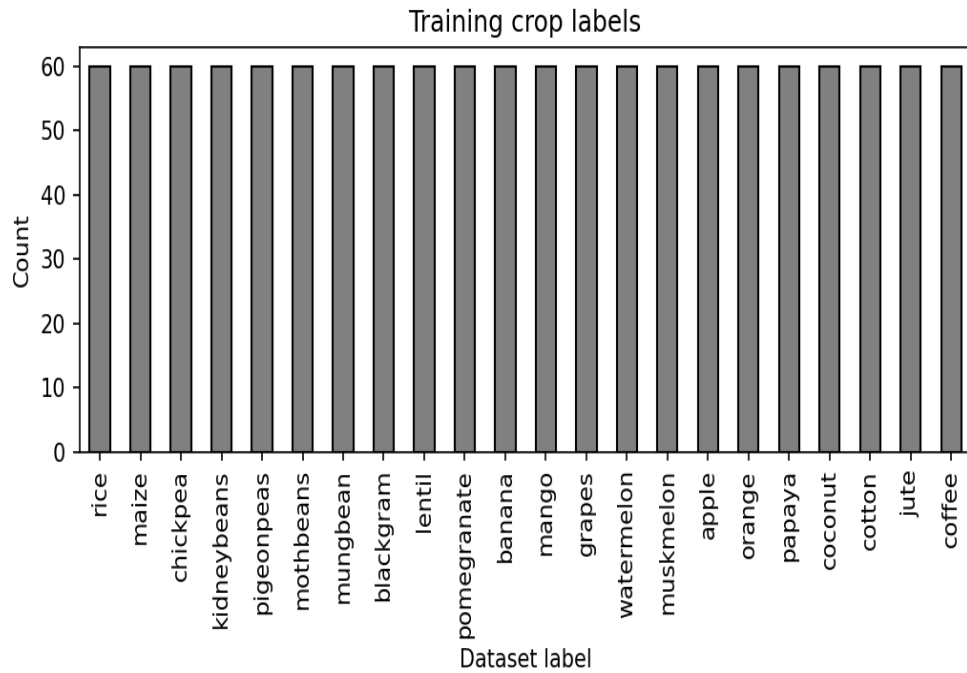


Figure 6. Training crop-label counts (balanced across 22 classes).

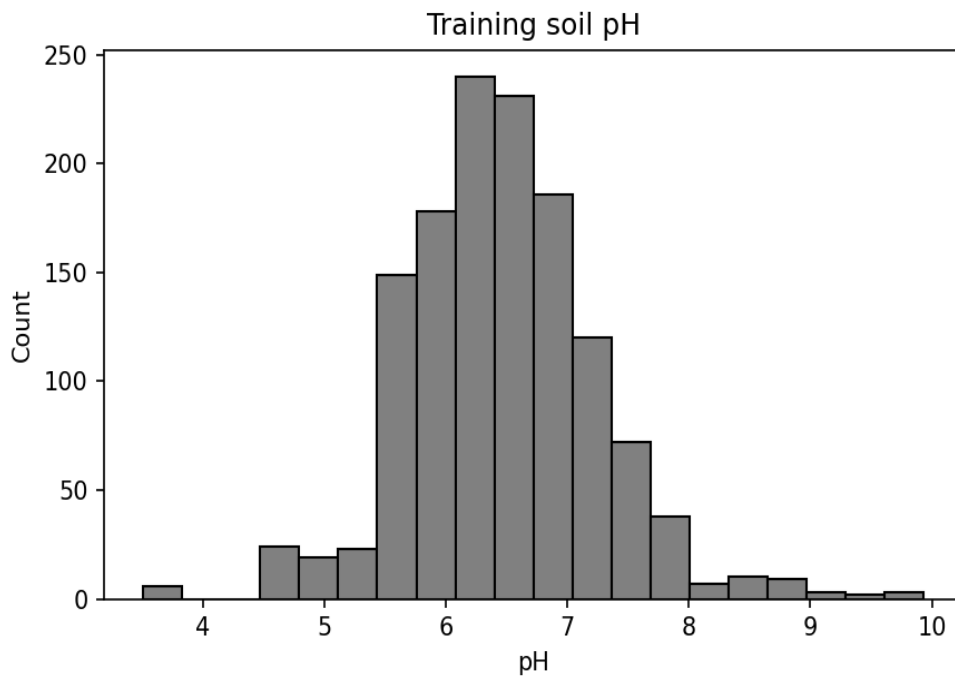


Figure 7. Training soil pH distribution.

The training soil pH distribution for the crop-label classification extension displays a unimodal, near-normal profile centered around a mean of 6.46 and a median of 6.43 with standard deviation $\sigma = 0.78$. The bulk of agricultural observations lie within the moderately acidic to neutral range (5.5 to 7.5), which represents the agronomically viable envelope for the majority of the 22 target crops. Truncated tails stretching down to 3.50 and up to 9.94 capture specialized crops (such as tea/mothbeans in acidic soils or alkaline-tolerant varieties) without exhibiting extreme non-finite anomalies or negative invalid values.

Models and validation

Three candidates: a majority-class baseline; standardized multinomial logistic regression (max_iter=2000); and a 60-tree random forest (min_samples_leaf=2).

Model	Accuracy	Macro precision	Macro recall	Macro F1
forest	0.993	0.993	0.993	0.993
logistic	0.973	0.973	0.973	0.973
majority	0.045	0.002	0.045	0.004

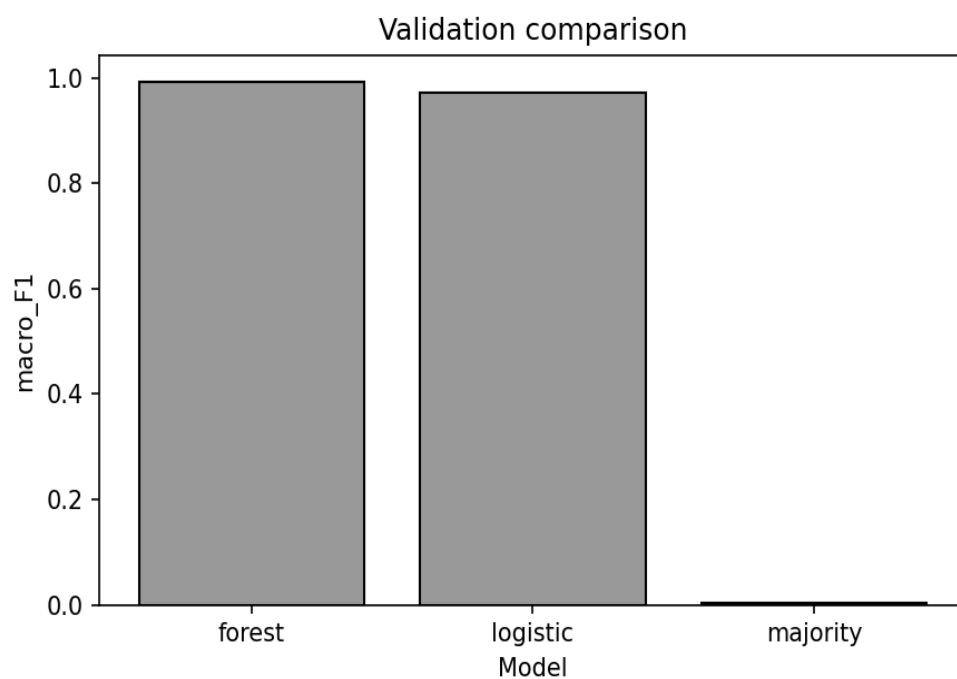


Figure 8. Validation macro-F1 comparison.

Selected model: forest (highest macro F1).

Locked test and confusion analysis

Model	Accuracy	Macro precision	Macro recall	Macro F1
forest (selected)	0.991	0.992	0.991	0.991
majority (predeclared reference)	0.045	0.002	0.045	0.004

Locked-test crop labels

True label	apple	banana	blackgram	chickpea	coconut	coffee	cotton	grapes	jute	kidneybeans	lentil	maize	mango	mothbeans	mungbean	muskmelon	orange	papaya	pigeonpeas	pomegranate	rice	watermelon
apple	20	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
banana	0	20	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
blackgram	0	0	20	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
chickpea	0	0	0	20	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
coconut	0	0	0	0	20	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
coffee	0	0	0	0	0	20	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
cotton	0	0	0	0	0	0	20	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
grapes	0	0	0	0	0	0	0	20	0	0	0	0	0	0	0	0	0	0	0	0	0	0
jute	0	0	0	0	0	0	0	0	20	0	0	0	0	0	0	0	0	0	0	0	0	0
kidneybeans	0	0	0	0	0	0	0	0	0	20	0	0	0	0	0	0	0	0	0	0	0	0
lentil	0	0	0	0	0	0	0	0	0	0	19	0	0	1	0	0	0	0	0	0	0	0
maize	0	0	0	0	0	0	0	0	0	0	0	20	0	0	0	0	0	0	0	0	0	0
mango	0	0	0	0	0	0	0	0	0	0	0	0	20	0	0	0	0	0	0	0	0	0
mothbeans	0	0	0	0	0	0	0	0	0	0	0	0	0	20	0	0	0	0	0	0	0	0
mungbean	0	0	0	0	0	0	0	0	0	0	0	0	0	0	20	0	0	0	0	0	0	0
muskmelon	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	20	0	0	0	0	0	0
orange	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	20	0	0	0	0	0
papaya	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	20	0	0	0	0
pigeonpeas	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	20	0	0	0
pomegranate	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	20	0	0
rice	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	17	0
watermelon	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	20
Predicted label	apple	banana	blackgram	chickpea	coconut	coffee	cotton	grapes	jute	kidneybeans	lentil	maize	mango	mothbeans	mungbean	muskmelon	orange	papaya	pigeonpeas	pomegranate	rice	watermelon

Figure 9. Locked-test confusion matrix (22 classes).

Which labels are confused, and why it matters: the only off-diagonal confusion in the locked test is between rice and jute — 3 of 20 rice cases were misclassified as jute. Both crops share the humid, high-rainfall growing-condition profile in this feature space (comparable temperature, humidity and rainfall ranges), so N/P/K/temperature/humidity/pH/rainfall alone cannot always separate them. Every other class was classified perfectly (20/20), which is expected given the balanced, synthetic-feeling nature of this benchmark; near-perfect accuracy on a compact, augmented dataset like D3 should not be read as evidence that a real deployed system would perform this well on messier field measurements.

Reminder from the manual: this classifier returns a dataset label, not a farming prescription. Confidence should not be read from raw class probabilities without separate calibration, and the label provenance for D3 (augmented/combined public data) is not fully documented, so per-class performance here should not be cited as agronomic evidence.

Known Limitations

- Fixed teaching hyperparameters (tree depth, forest size) were not tuned; they are not claimed to be optimal.
- No observed weather, soil, or irrigation predictors enter the core regression — only state, district, season and year — so single-year shocks (Section 9) cannot be anticipated by this model.
- District-level aggregation and a repeated-district panel mean the model may partly memorize district-level yield levels rather than learning a transferable spatial pattern; predictions for unseen districts are out of scope and are rejected by the inference-time category check.
- Only 23 years and one state (West Bengal) were used for the core, smaller than a full multi-state D1/D2 export; conclusions are scoped to this region and should not be generalized nationally without re-running on additional states.
- Rolling-origin and two-year robustness results are descriptive stability checks, not confidence intervals; three or two points are too few for a valid interval estimate.
- Feature importance in the fitted forest describes association within this model, not a causal explanation of what drives yield.
- The classification extension's near-perfect scores reflect a compact, augmented benchmark dataset with undocumented label-generation provenance, and do not establish real-world crop-recommendation accuracy.
- This is a retrospective demonstration exercise; no operational deployment, real-time monitoring, or agronomic decision system is implied or supported by these results.

Responsible Agricultural Analytics

The data used are aggregate, publicly sourced district-season-year statistics with no personal or farm-level identifiers, so no additional consent or coarsening step was required for this exercise. Representation was checked at the district level (22 districts across 3 seasons and 23 years); no attempt was made to draw personal or social-group fairness conclusions from regional error differences, which reflect agricultural and geographic factors (e.g., hill vs. plains terrain) rather than any property of individuals.

Neither the yield regression nor the crop-label classifier should be used, on their own, to make a real resource-allocation or crop-selection decision. A real system would additionally need soil testing, irrigation availability, crop variety information, economic constraints, weather forecasts and local expert judgment, none of which are present in either dataset used here. Predictions are estimates conditioned on the available historical pattern, not facts about a specific future harvest.

Conclusion

For Rice yield prediction across West Bengal districts (1997–2019), the random forest achieved the lowest validation MAE (0.258 t/ha) among four candidates and locked-test MAE of 0.364 t/ha ($R^2 = 0.402$), improving on the median baseline's test MAE of 0.617 t/ha. Compared on the same validation split, both ridge regression (0.281) and the decision tree (0.308) trailed the forest, though the gap to ridge narrowed and reversed at one of the three development-year rolling origins, suggesting ridge's linear year-trend extrapolation becomes competitive as more years accumulate. The largest failures occurred in a hill district (Kalimpong, recurring across two winters) and in two apparent single-year shocks (Purba Bardhaman and Medinipur East, both coastal/flood-exposed areas), plus one high-yield outlier (Maldah) that the forest's leaf-averaging compressed toward the mean. These results support a defensible, reproducible comparison of simple and nonlinear models for a bounded, chronologically evaluated prediction task, but do not establish a validated operational yield-forecasting system, a causal account of what drives West Bengal rice yield, or any claim about optimal crop selection.

For the crop-label classification extension, the random forest reached 0.991 test accuracy and 0.991 macro F1 on a held-out stratified split of the balanced 22-class D3-style benchmark, with the only confusion occurring between rice and jute. This supports the narrow claim that N/P/K/temperature/humidity/pH/rainfall are highly informative for distinguishing this dataset's crop labels, but does not establish real-world crop-recommendation accuracy given the benchmark's undocumented augmentation and label-generation provenance.

GitHub Repository: <https://github.com/selvavinayagamkamaraj/MDI3003-Lab08-CropYield-23MID0328.git>

Discussion Questions

2. Why does a production predictor invalidate a yield-derived target?

Because `yield_t_ha` is calculated as production divided by area in the source data, giving the model production (with area, or alone combined with a correlated area proxy) lets it reconstruct the target almost exactly rather than learn a genuine location/time pattern. The validation and test scores would then measure how well the model can invert arithmetic, not how well it predicts an unobserved future harvest — precisely the leakage the manual's REG whitelist is designed to prevent.

4. Why might a forest fail to beat the median?

A forest can fail to beat a median baseline when the signal in the available features is very weak relative to noise, when the training set is too small or too homogeneous for bootstrap-based variance reduction to help, or when the target is nearly constant within the evaluation subgroup (making the median already near-optimal). In this experiment the forest did beat the median comfortably (validation MAE 0.258 vs. 0.520), because state/district/season/year carry real, learnable location- and season structure in West Bengal rice yields; a dataset with less structured categorical signal could easily reverse this.

5. What does a negative R^2 tell you about this evaluation set?

A negative R^2 means the model's squared error is larger than the squared error of simply predicting the evaluation set's own mean for every row — i.e., the model performs worse than a trivial constant predictor on that specific set. The median baseline's negative R^2 on both the validation and test sets (Sections 6–7) reflects that a single constant cannot capture the district-to-district and season-to-season spread in rice yield, which is expected and simply motivates comparing it against the trained models rather than treating R^2 alone as a pass/fail threshold.

7. Why does chronological splitting not prove spatial generalization?

The chronological split (train on 1997–2015, validate on 2016–2017, test on 2018–2019) only holds out later years; the same 22 districts reappear in every split. It therefore tests whether the model generalizes forward in time for districts it has already seen, not whether it generalizes to unseen districts or regions. A genuinely spatial generalization claim would require the held-out spatial-shift design described in the manual's Extension 4 (hold out entire districts), which was not run here.

11. Why are two test years insufficient for a strong temporal confidence interval?

A confidence interval needs enough independent-ish observations of the estimator's variability to characterize its sampling distribution. Two years give only two point estimates of test-year MAE (0.358 and 0.370 t/ha in Section 8.1), which is enough to say the two years look similar but far too few to fit a distribution, estimate a standard error, or bound uncertainty with any statistical rigor — hence the manual's framing of Section 8 as descriptive robustness rather than a valid interval.

13. How can apparently balance augmented data still be biased?

The D3 crop-recommendation dataset used in Section 11 has exactly 100 rows per class, which looks perfectly balanced, but balance in row counts does not guarantee that the underlying feature distributions (N, P, K, temperature, humidity, pH, rainfall) were sampled the way real farms in each crop's growing regions are actually distributed. If the data were generated or augmented from a small number of reference profiles per crop, the classifier could be learning to separate a handful of synthetic clusters rather than the true diversity of real-world soil/weather conditions for each crop — which is why the manual repeatedly cautions against treating D3's near-perfect accuracy as agronomic evidence.

Common Mistakes Checklist

Mistake	
Use production as input	REG whitelist = [state, district, season, year] only; production/area excluded from every model
Mix kg/ha and t/ha	Source yield_t_ha verified as t/ha before use; no unit conversion mixing
Mix cotton bales with rice tonnes	Single-crop scope: crop == 'Rice' enforced by an explicit check in load_data
Randomly split a future-year task	Ordered year blocks: train < validation < test, asserted programmatically
Impute before splitting	SimpleImputer and StandardScaler are fit only inside the training-only pipeline
Tune on test residuals	TEST_LOCK written before scoring; test opened exactly once per task
Delete repeated keys blindly	Duplicate-key check found zero repeated state-district-season-year keys; none deleted
Report only accuracy (classification)	Macro precision/recall/F1 and a full confusion matrix reported, not accuracy alone
Treat labels as expert advice	Section 11.3 explicitly states the classifier returns a dataset label, not a farming prescription
Call feature importance causal	Section 12 explicitly limits importance to association within the fitted model
Load an untrusted serialized model	Only locally trained .joblib artifacts from this run were loaded